

1 (a) (i) Write the following simultaneous equations as a matrix equation.

$$x + y + 2z = 7$$

$$2x - 4y - 3z = -5$$

$$-5x + 3y + 5z = 13$$

[1]

(ii) Hence solve the equations.

[2]

(b) Determine the set of values of the constant k for which the matrix equation

$$\begin{pmatrix} k+1 & 1 \\ 2 & k \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 23 \\ -17 \end{pmatrix}$$

has a unique solution.

[3]